

TECHNICAL ARCHITECTURE — PHASE 1

Technical Architecture — Phase 1

The Archive MVP

v2.0 · Richard Uzor — May 2026 · Devstrike Digital Limited

01 — PHASE 1 ARCHITECTURE PRINCIPLE

Phase 1 architecture is intentionally minimal. One Next.js app on Vercel, one Node.js API on Railway, one PostgreSQL database. No cache layer, no job queue, no WebSockets. All of these can be added later — starting without them keeps Phase 1 fast to build and easy to debug.

02 — PHASE 1 STACK

Component	Technology	Notes
Frontend	Next.js 14 App Router	SSR for all pages. No CSR pages in Phase 1.
API	Node.js + TypeScript + Express	REST. JSON. JWT auth in httpOnly cookies.

Component	Technology	Notes
Database	PostgreSQL 15 via Railway	Prisma ORM. Migrations committed to repo.
Storage	Cloudflare R2	Pastor/church images, generated quote cards.
Email	Resend (free tier)	Verification emails, Q&A notifications.
Analytics	Plausible	Privacy-first. No cookies. No consent banner.
Error tracking	Sentry (free tier)	Frontend + backend exceptions.
Deployment	Vercel (Hobby) + Railway (Starter)	Both free. GitHub-integrated CI/CD.

03 — PHASE 1 API ENDPOINTS

Method	Endpoint	Auth	Description
GET	/api/v1/messages	Public	List messages — category, pastor, church, page filters
POST	/api/v1/messages	Public*	Create message — *guest requires KYC body fields
GET	/api/v1/messages/:id	Public	Message detail with Q&A thread
POST	/api/v1/messages/:id/react	Public	Add reaction
POST	/api/v1/messages/:id/report	Public	Report with reason
POST	/api/v1/messages/:id/question	Public	Ask a question
GET	/api/v1/pastors	Public	List pastors with filters
POST	/api/v1/pastors	Account	Create pastor profile
GET	/api/v1/pastors/:id	Public	Pastor profile + messages

Method	Endpoint	Auth	Description
PUT	/api/v1/pastors/:id	Account	Update mutable fields
GET	/api/v1/churches	Public	List churches with filters
POST	/api/v1/churches	Account	Create church profile
GET	/api/v1/churches/:id	Public	Church profile + pastors + messages
GET	/api/v1/events	Public	List events — type, church filters
POST	/api/v1/events	Account	Create event
GET	/api/v1/events/:id	Public	Event detail + messages + testimonies
POST	/api/v1/events/:id/testimony	Public	Submit testimony
GET	/api/v1/search	Public	Search messages, pastors, churches
POST	/api/v1/auth/register	Public	Create account
POST	/api/v1/auth/login	Public	Login — returns JWT
GET	/api/v1/users/me	Account	Authenticated user profile
GET	/api/v1/users/me/saved	Account	Saved messages and events

04 — QUOTE CARD GENERATION

The shareable quote card is the most important Phase 1 feature technically. Generated client-side using html2canvas with server-side fallback via Satori (headless canvas rendering).

Aspect	Spec
Canvas size	1080 × 1080px (Instagram square / WhatsApp-friendly)
Background	Deep Navy (#0F1B2D)
Quote font	DM Serif Display, 36px, white

Aspect	Spec
Attribution	Inter, 18px, Sacred Gold — Pastor Name · Church · Date
Logo	WaMPaS wordmark, top-left, 24px
Opening quote mark	DM Serif Display, 96px, gold — decorative
Generation trigger	User taps Share on message detail page
Storage	Generated PNG stored in Cloudflare R2 with 24-hour TTL